New CP — Service-Instance API (F07 + Billability + Q-B8..15 Approved Clarifications)

OpenAPI 3.0.3 | Version 1.0.0

Service-instance API contract for New CP, merging the foundation health probe, F07 per-item billed party target schema, Connect create-service fields, CP internal clerk editing controls, full pricing breakdown, V3 database schema validation, and the Technical Lead approved Q-B8 to Q-B15 clarifications (24 October 2026: aggregate ETag/If-Match optimistic concurrency, atomic required hold details on workflow transition, authorized internal price override and zero-charge reasons, canonical field types and typed conditional rules, demo quantity default, and independent billability alongside F07 billed party). The migration and ETL are supplied as implementation artifacts; this document does NOT certify V204 deployment, dump-specific payer mapping, stage-90 reconciliation, downstream finance updates, tenant catalog mappings or live integration tests.

Base Server URL	http://localhost:3000 (Local New CP backend; URL example, not a verified deployment)
Authentication	HTTP Bearer Token (BearerAuth — JWT or opaque token per Google Auth adapter)
Contract Status	APPROVED_QB8_QB15_IMPLEMENTATION_TARGET; F07_AND_LIVE_API_UNVERIFIED; NOT_FINAL
F07 Target Mapping	Column: orders.order_items.billed_to_contact_id | Origin: orders.order_items.billed_party_origin
Independent Billability	Column: orders.order_items.is_billable (V011 existing NOT NULL DEFAULT true; independent of F07 V204)
Concurrency & Locks	Aggregate ETag + required If-Match (412 stale / 428 missing); financial lock after any charge or invoice

1. API Endpoints Overview

Method	Endpoint Path	Tag	Operation ID & Summary
GET	`/health`	System	getHealth: Liveness/readiness probe
GET	`/api/v1/catalog/services`	Catalog	listDemoCatalogServices: List authorized sellable service options (tenant scoped)
GET	`/api/v1/catalog/services/{serviceUuid}/form-schema`	Catalog	getServiceFormSchema: Get actual tenant catalog field and condition metadata
GET	`/api/v1/orders/{orderUuid}/services`	Service Instances	listOrderServiceItems: List independent service instances on an authorized order
POST	`/api/v1/orders/{orderUuid}/services`	Service Instances	createOrderServiceItem: Create a new independent service instance on an order
GET	`/api/v1/services/{serviceItemUuid}`	Service Instances	getServiceItem: Get a single service instance with ETag aggregate revision
PATCH	`/api/v1/services/{serviceItemUuid}`	Service Instances	patchServiceItem: Update a service instance with If-Match concurrency guard
GET	`/api/v1/services/{serviceItemUuid}/available-transitions`	Workflow	listServiceTransitions: List currently authorized manual workflow transitions
POST	`/api/v1/services/{serviceItemUuid}/transitions`	Workflow	advanceServiceWorkflow: Execute an atomic workflow state transition with If-Match guard

Method	Endpoint Path	Tag	Operation ID & Summary
GET	`/api/v1/services/{serviceItemUuid}/history`	Workflow	getServiceHistory: Get status and workflow transition history for a service instance
GET	`/api/v1/orders/{orderUuid}/eligible-billed-parties`	Service Instances	listEligibleOrderItemBilledParties: List eligible service-level billed parties for one authorized order
GET	`/api/v1/catalog/hold-reasons`	Catalog, Lookups	listEligibleHoldReasons: Return configured hold reasons; temporary synthetic choices are nonproduction only
GET	`/api/v1/lookups/zero-charge-reasons`	Lookups	listZeroChargeReasons: List active zero charge reasons
GET	`/api/v1/lookups/hold-reasons`	Lookups	listHoldReasons: List active hold reasons (lookup alias for /api/v1/catalog/hold-reasons)

2. Detailed Endpoint Specifications

[GET] /health

Operation ID: getHealth | Tag: System | Summary: Liveness/readiness probe

Responses:

Status	Description & Headers	Response Schema
200	Service is healthy	`object {status: string}`

[GET] /api/v1/catalog/services

Operation ID: listDemoCatalogServices | Tag: Catalog | Summary: List authorized sellable service options (tenant scoped)

Parameters:

Name	In	Required	Type	Description
`categoryCode`	query	No	`string`	-

Responses:

Status	Description & Headers	Response Schema
200	Success	`array<Ref: CatalogService>`
400	Bad request / invalid field value	`Ref: Problem`
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	`Ref: Problem`

[GET] /api/v1/catalog/services/{serviceUuid}/form-schema

Operation ID: getServiceFormSchema | Tag: Catalog | Summary: Get actual tenant catalog field and condition metadata

Parameters:

Name	In	Required	Type	Description
`serviceUuid`	path	**Yes**	`string (uuid)`	-

Responses:

Status	Description & Headers	Response Schema
200	Success	`Ref: FormDefinition`
400	Bad request / invalid field value	`Ref: Problem`
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	`Ref: Problem`

[GET] /api/v1/orders/{orderUuid}/services

Operation ID: listOrderServiceItems | Tag: Service Instances | Summary: List independent service instances on an authorized order

Parameters:

Name	In	Required	Type	Description
`orderUuid`	path	**Yes**	`string (uuid)`	-

Responses:

Status	Description & Headers	Response Schema
200	Success	`array<Ref: ServiceItem>`
400	Bad request / invalid field value	`Ref: Problem`
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	`Ref: Problem`

[POST] /api/v1/orders/{orderUuid}/services

Operation ID: createOrderServiceItem | Tag: Service Instances | Summary: Create a new independent service instance on an order

Create a new independent service-instance UUID even when same sellable item already exists. Server enforces field rules, authorization and tenant-scoped transaction. Idempotency-Key proposal requires explicit policy confirmation; intentional second submissions use distinct keys.

Parameters:

Name	In	Required	Type	Description
`orderUuid`	path	**Yes**	`string (uuid)`	-
`Idempotency-Key`	header	No	`string`	Proposed retry protection; exact retention/replay policy is a technical decision.

Request Body (application/json): Ref: CreateServiceItem

Responses:

Status	Description & Headers	Response Schema
201	Created service item • Header ETag: Initial opaque revision of the created service aggregate.	`Ref: ServiceItem`
400	Bad request / invalid field value	`Ref: Problem`
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	`Ref: Problem`

[GET] /api/v1/services/{serviceItemUuid}

Operation ID: getServiceItem | Tag: Service Instances | Summary: Get a single service instance with ETag aggregate revision

Return ETag covering the entire editable service state. Implementation must update revision on child field/workflow writes, not only order_items.updated_at.

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	`string (uuid)`	-

Responses:

Status	Description & Headers	Response Schema
200	Success • Header ETag: Opaque revision of the entire editable service aggregate (including fields/workflow). Store and send on next write.	`Ref: ServiceItem`
400	Bad request / invalid field value	`Ref: Problem`
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	`Ref: Problem`

[PATCH] /api/v1/services/{serviceItemUuid}

Operation ID: patchServiceItem | Tag: Service Instances | Summary: Update a service instance with If-Match concurrency guard

Requires If-Match; approved CP authorized unitPrice overrides must not be silently replaced with contract rate. Preserve zero-charge reason required at final line_total zero and financial locks.

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	`string (uuid)`	-
`If-Match`	header	**Yes**	`string`	ETag of loaded service aggregate. Required on mutations. 412 if stale, 428 if absent.

Request Body (application/json): Ref: PatchServiceItem

Responses:

Status	Description & Headers	Response Schema
200	Success • Header ETag: Opaque revision of the entire editable service aggregate (including fields/workflow). Store and send on next write.	`Ref: ServiceItem`
400	Bad request / invalid field value	`Ref: Problem`
401	Not authenticated	`-`
403	Denied	`-`
404	Not found or masked by tenant access	`-`
409	Business state conflict	`-`
412	Stale version: changed by another user, no write occurred	`Ref: Problem`
422	Validation error	`Ref: Problem`
428	Missing required If-Match precondition	`Ref: Problem`

[GET] /api/v1/services/{serviceItemUuid}/available-transitions

Operation ID: listServiceTransitions | Tag: Workflow | Summary: List currently authorized manual workflow transitions

Return currently authorized manual transitions from selected workflow instance. Do not hardcode legacy numeric statuses.

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	`string (uuid)`	-

Responses:

Status	Description & Headers	Response Schema
200	Success	`array<Ref: AvailableTransition>`
400	Bad request / invalid field value	`Ref: Problem`
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	`Ref: Problem`

[POST] /api/v1/services/{serviceItemUuid}/transitions

Operation ID: advanceServiceWorkflow | Tag: Workflow | Summary: Execute an atomic workflow state transition with If-Match guard

Requires If-Match; when destination is on_hold, mandatory holdType + holdReasonCode; commit service status, hold details and history in one DB transaction. Temporary lookup reason choices are for authorized test/demo environments only.

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	`string (uuid)`	-
`If-Match`	header	**Yes**	`string`	ETag of loaded service aggregate. Required on mutations. 412 if stale, 428 if absent.

Request Body (application/json): Ref: TransitionRequest

Responses:

Status	Description & Headers	Response Schema
200	Success • Header ETag: New aggregate version after atomic transition	`Ref: TransitionResult`
400	Bad request / invalid field value	`Ref: Problem`
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
412	Stale version: changed by another user, no write occurred	`Ref: Problem`
422	Validation error	`Ref: Problem`
428	Missing required If-Match precondition	`Ref: Problem`

[GET] /api/v1/services/{serviceItemUuid}/history

Operation ID: getServiceHistory | Tag: Workflow | Summary: Get status and workflow transition history for a service instance

Parameters:

Name	In	Required	Type	Description
`serviceItemUuid`	path	**Yes**	`string (uuid)`	-

Responses:

Status	Description & Headers	Response Schema
200	Success	`array<Ref: HistoryEvent>`
400	Bad request / invalid field value	`Ref: Problem`
401	Not authenticated	-
403	Denied	-
404	Not found or masked by tenant access	-
409	Business state conflict	-
422	Validation error	`Ref: Problem`

[GET] /api/v1/orders/{orderUuid}/eligible-billed-parties

Operation ID: listEligibleOrderItemBilledParties | Tag: Service Instances | Summary: List eligible service-level billed parties for one authorized order

Resolve order Doctor/default and Scanning Center within tenant and staff permission, return contact UUIDs for selection; do not return unrestricted contacts. Endpoint proposal until backend contract review.

Parameters:

Name	In	Required	Type	Description
`orderUuid`	path	**Yes**	`string (uuid)`	-

Responses:

Status	Description & Headers	Response Schema
200	Eligible billed parties	`array<Ref: BilledPartyChoice>`
401	Not authenticated	-
403	Denied	-
404	Order not found or tenant-masked	-

[GET] /api/v1/catalog/hold-reasons

Operation ID: listEligibleHoldReasons | Tag: Catalog, Lookups | Summary: Return configured hold reasons; temporary synthetic choices are nonproduction only

Returns valid hold reasons filtered by holdType (customer or internal) for atomic on_hold workflow transitions (Q-B9).

Parameters:

Name	In	Required	Type	Description
`holdType`	query	**Yes**	`Ref: HoldType`	Required hold type filter (customer or internal)

Responses:

Status	Description & Headers	Response Schema
200	Configured hold reasons	`array<Ref: HoldReasonOption>`
401	Not authenticated	-
403	Denied	-

[GET] /api/v1/lookups/zero-charge-reasons

Operation ID: listZeroChargeReasons | Tag: Lookups | Summary: List active zero charge reasons

Returns active reasons required when an order item line total is $0.00 to satisfy DB constraint ck_order_items__zero_charge_has_reason (Q-B10).

Responses:

Status	Description & Headers	Response Schema
200	Success	`array<Ref: ZeroChargeReason>`
401	Not authenticated	-
403	Denied	-

[GET] /api/v1/lookups/hold-reasons

Operation ID: listHoldReasons | Tag: Lookups | Summary: List active hold reasons (lookup alias for /api/v1/catalog/hold-reasons)

Returns valid reasons when placing an item on hold with customer or internal hold type; temporary synthetic choices are nonproduction only.

Parameters:

Name	In	Required	Type	Description
`holdType`	query	No	`Ref: HoldType`	Filter reasons by hold type (customer or internal)

Responses:

Status	Description & Headers	Response Schema
200	Success	`array<Ref: HoldReasonOption>`
401	Not authenticated	-
403	Denied	-

3. Component Schemas (Data Models)

Schema: Problem (object)

Property	Type & Constraints	Req	Description
`code`	`string`	**Yes**	-
`message`	`string`	**Yes**	-
`requestId`	`string`	No	-
`details`	`object {field: string, reason: string}`	No	For ZERO_CHARGE_REASON_REQUIRED (422), details.field must equal zeroChargeReasonCode; other errors may include structured code-specific detail. (Sub-properties: field, reason)

Schema: FormOption (object)

Property	Type & Constraints	Req	Description
`code`	`string`	**Yes**	-
`label`	`string`	**Yes**	-
`active`	`boolean`	No	-

Schema: CatalogService (object)

Property	Type & Constraints	Req	Description
`uuid`	`string (uuid)`	**Yes**	-
`code`	`string`	**Yes**	-
`name`	`string`	**Yes**	-
`itemKind`	`string [enum: service]`	**Yes**	-
`isDigital`	`boolean`	No	-
`workflowCode`	`string (nullable)`	No	-

Schema: CanonicalFieldType (string)

Data types allowed by V3 catalog.field_types.data_type (Q-B11), verify actual seeded code values; UI widgets such as radio vs select are presentation-only.

`Allowed Enum Values: text, number, boolean, date, single_select, multi_select, file, tooth_chart`

Schema: FormValidation (object)

Known V3 validation keys (Q-B12); backend must review actual existing JSON configurations before freezing full key vocabulary.

Property	Type & Constraints	Req	Description
`min`	`number`	No	-
`max`	`number`	No	-
`max_length`	`integer [min: 0]`	No	-
`pattern`	`string`	No	-

Schema: FieldCondition (oneOf)

`Polymorphic Variants (oneOf): Ref: FieldConditionLeaf, Ref: FieldConditionAll, Ref: FieldConditionAny, Ref: FieldConditionNot`

Schema: FieldConditionLeaf (object)

Exactly one of field/attribute; field codes resolved using tenant catalog; value vs values depends on op; enforce with application validator.

Property	Type & Constraints	Req	Description
`field`	`string`	No	-
`attribute`	`string`	No	-
`op`	`string [enum: eq, ne, in, not_in, is_set, is_not_set, gt, gte, lt, lte]`	**Yes**	-
`value`	`any`	No	-
`values`	`array<any>`	No	-

Schema: FieldConditionAll (object)

Property	Type & Constraints	Req	Description
`all`	`array<Ref: FieldCondition>`	**Yes**	-

Schema: FieldConditionAny (object)

Property	Type & Constraints	Req	Description
`any`	`array<Ref: FieldCondition>`	**Yes**	-

Schema: FieldConditionNot (object)

Property	Type & Constraints	Req	Description
not	Ref: FieldCondition	**Yes**	-

Schema: ConditionalFieldRule (object)

Rule kind, ordered options precedence and conditional expression V3-derived (Q-B12); server evaluator authoritative.

Property	Type & Constraints	Req	Description
ruleKind	string [enum: visible, required, options]	**Yes**	-
condition	Ref: FieldCondition	**Yes**	-
optionCodes	array<string>	No	-
sortOrder	integer	No	-
active	boolean	No	-

Schema: FormField (object)

Property	Type & Constraints	Req	Description
uuid	string (uuid)	**Yes**	-
code	string	**Yes**	-
label	string	**Yes**	-
type	Ref: CanonicalFieldType	**Yes**	-
required	boolean	**Yes**	-
defaultValue	string (nullable)	No	-
validation	Ref: FormValidation	No	-
conditionalRules	array<Ref: ConditionalFieldRule>	No	-
options	array<Ref: FormOption>	**Yes**	-

Schema: FormDefinition (object)

Property	Type & Constraints	Req	Description
service	Ref: CatalogService	**Yes**	-
fields	array<Ref: FormField>	**Yes**	-
evidence	string	No	Indicates whether tenant runtime mapping is verified

Schema: FormAnswer (object)

Form answer model supporting both the clean polymorphic value + optionCodes array (Section 4 Strategy 1 & Section 5 Approach A) and the v0.9.2 shared client typed carriers. Validation is performed from tenant catalog + source-evidenced rules, including conditional requiredness.

Property	Type & Constraints	Req	Description
`fieldCode`	`string`	**Yes**	Stable machine code of the form question (from catalog.fields.code)
`value`	`oneOf: string | number | boolean`	No	Natural polymorphic answer value for free-form inputs: string (text or ISO date), number, or boolean. Backend validates and maps to appropriate SQL column.
`optionCodes`	`array<string>`	No	Unified option codes array for choice-based fields: single-select fields supply 1 element e.g. ["FORMLABS_3B"], multi-select fields supply multiple elements e.g. ["UPPER", "LOWER"], and unselected/cleared fields supply [].
`optionCode`	`string`	No	Compatibility single-select option code (used by v0.9.2 shared TS client); backend normalizes with optionCodes.
`valueText`	`string`	No	Compatibility text carrier (used by v0.9.2 shared TS client).
`valueNumber`	`number`	No	Compatibility numeric carrier (used by v0.9.2 shared TS client).
`valueBoolean`	`boolean`	No	Compatibility boolean carrier (used by v0.9.2 shared TS client).
`valueDate`	`string (date)`	No	Compatibility date carrier (used by v0.9.2 shared TS client).

Schema: CreateServiceItem (object)

Service-level payer is backed by F07 V204 orders.order_items.billed_to_contact_id. Application resolves billedToContactUuid to crm.contacts.id under tenant-scoped transaction. Omit to select order default once; do not allow client to set provenance. Financial writes require runtime F07 gates. Every service instance independently retains billability and intended payer.

Property	Type & Constraints	Req	Description
`sellableItemUuid`	`string (uuid)`	**Yes**	-
`answers`	`array<Ref: FormAnswer>`	**Yes**	-
`isRedo`	`boolean [default: False]`	No	-
`redoOfItemUuid`	`string (uuid)`	No	-
`serviceNotes`	`string`	No	-
`billedToContactUuid`	`string (uuid)`	No	Optional selected payer for THIS item; when omitted creation copies order default ONCE. Must be eligible for the specific order and tenant.
`isBillable`	`boolean [default: True]`	No	Independent service-instance billability (Q-B14); true = chargeable/billable, false = never invoiced per V3 column definition. Not the Doctor/Scanning Center payer selection; never map legacy toBeBilled to this flag. Changes require clerk permission and are financially locked once charged/invoiced.
`unitPrice`	`string`	No	Authorized CP-adjustable per-unit monetary price, decimal string (Q-B10). Backend supplies contract-based initial value, then accepts permitted clerk override without silently re-pricing it to contract amount. Enforce currency scale, audit and financial locks; do not trust unprivileged writers.
`quantity`	`number [default: 1] [min: 0.001] [step: 0.001]`	No	Selected demo workflows default to 1 (Q-B13). Do NOT interpret as universal integer-only or minimum 1: V3 supports >0 and up to 3 decimals (numeric(10,3)); item-specific policy pending.
`zeroChargeReasonCode`	`string`	No	Lookup code resolved server-side (Q-B10); mandatory for new lines whose final line_total equals zero, including discount/voucher, to satisfy DB constraint ck_order_items__zero_charge_has_reason.
`zeroChargeNote`	`string [maxLen: 255]`	No	Free-text explanation when line_total is zero (max 255 chars).

Schema: PatchServiceItem (object)

Do not accept caller-supplied billedPartyOrigin. Reject per-item payer changes once a charge, invoice association or invoice line exists. Other monetary changes obey their own permissions and V3 financial locks. Never expose arbitrary status or hold changes here instead of configured workflow actions (POST /transitions). Every service instance independently retains billability and intended payer.

Property	Type & Constraints	Req	Description
`answers`	`array<Ref: FormAnswer>`	No	-
`isRedo`	`boolean`	No	-
`redoOfItemUuid`	`string (uuid)`	No	-
`serviceNotes`	`string`	No	-
`pendingReason`	`string`	No	-
`billedToContactUuid`	`string (uuid)`	No	Change intended payer for this single service instance only before any financial event; server writes billed_party_origin=user_selected and audits the change.
`isBillable`	`boolean`	No	Independent service-instance billability (Q-B14); true = chargeable/billable, false = never invoiced per V3 column definition. Not the Doctor/Scanning Center payer selection; never map legacy toBeBilled to this flag. Changes require clerk permission and are financially locked once charged/invoiced. This property changes billability only, not billedToContactUuid.
`unitPrice`	`string`	No	Authorized CP-adjustable per-unit monetary price, decimal string (Q-B10). Backend supplies contract-based initial value, then accepts permitted clerk override without silently re-pricing it to contract amount. Enforce currency scale, audit and financial locks; do not trust unprivileged writers.
`quantity`	`number [min: 0.001] [step: 0.001]`	No	Demo currently defaults to 1 (Q-B13); per-service validation pending; V3 permits positive decimals up to 3 places.
`zeroChargeReasonCode`	`string (nullable)`	No	Lookup code resolved server-side (Q-B10); mandatory when final line_total equals zero, including discount/voucher. Can be omitted or nullified if final total is nonzero.
`zeroChargeNote`	`string [maxLen: 255]`	No	-

Schema: ServiceItem (object)

Each service instance has independent item-level billed party, billability, pricing breakdown, form answers, status and history.

Property	Type & Constraints	Req	Description
`uuid`	`string (uuid)`	**Yes**	-
`orderUuid`	`string (uuid)`	**Yes**	-
`sellableItemUuid`	`string (uuid)`	**Yes**	-
`lineNumber`	`integer [min: 1]`	No	-

Property	Type & Constraints	Req	Description
status	string	**Yes**	-
isRedo	boolean	No	-
redoOfItemUuid	string (uuid) (nullable)	No	UUID of parent item this line remakes
isInvoiced	boolean	No	-
currencyCode	string [maxLen: 3]	No	-
unitPrice	string	No	Decimal string, not floating point
fieldsAmount	string	No	Sum of answer surcharges recalculated by trigger trg_order_item_field_values__refresh_item
discountAmount	string	No	Applied discount amount
voucherAmount	string	No	Applied voucher amount
lineTotal	string	No	Final line amount (unitPrice * quantity + fieldsAmount - discountAmount - voucherAmount)
canEditPrice	boolean	No	Backend-derived permission hint for authorized CP clerk (Q-B10); API/DB enforce.
answers	array<Ref: FormAnswer>	**Yes**	-
serviceNotes	string	No	-
billedToContactUuid	string (uuid)	**Yes**	Authoritative item-level intended payer (crm.contacts.uuid), not orders.orders.contact_id or charges.contact_id.
billedPartyOrigin	Ref: BilledPartyOrigin	**Yes**	-
canEditBilledParty	boolean	No	Server-calculated permission and financial-lock hint; backend always enforces mutation rules.
isBillable	boolean [default: True]	**Yes**	Independent service-instance billability (Q-B14); true = chargeable/billable, false = never invoiced per V3 column definition. Read persisted orders.order_items.is_billable, independent from billedToContactUuid.
canEditBillability	boolean	No	Server-derived hint for authorized internal clerk (Q-B14); actual backend validation and financial lock remain mandatory.
quantity	number [default: 1] [min: 0.001] [step: 0.001]	No	Stored V3 quantity (numeric(10,3)); demo default 1 is not a general integer restriction (Q-B13).
pendingReason	string (nullable)	No	Reason the item is pending information
holdType	Ref: HoldType (nullable)	No	Active hold type when status is on_hold
holdReasonCode	string (nullable)	No	Active hold reason code when status is on_hold
zeroChargeReasonCode	string (nullable)	No	Code explaining why line_total is zero
zeroChargeNote	string (nullable)	No	-
etag	string	No	Optional body convenience only (Q-B8); authoritative version is ETag response header. Backend must version entire editable aggregate.

Property	Type & Constraints	Req	Description
createdAt	string (date-time)	No	-
updatedAt	string (date-time)	No	-

Schema: AvailableTransition (object)

Property	Type & Constraints	Req	Description
transitionId	integer (int64)	**Yes**	-
fromCode	string	**Yes**	-
toCode	string	**Yes**	-
label	string	No	-
requiresCapability	string	No	-
guard	object	No	-

Schema: TransitionRequest (object)

One atomic transition (Q-B9): when destination is on_hold, holdType and holdReasonCode are required; set status, hold type/reason, and workflow history together. Leaving hold clears active details while retaining history. Server validates against actual transition target.

Property	Type & Constraints	Req	Description
transitionId	integer (int64)	**Yes**	-
comment	string	No	-
holdType	Ref: HoldType	No	-
holdReasonCode	string	No	Required alongside holdType for EVERY new transition into on_hold (Q-B9), resolved to orders.order_items.hold_reason_id from eligible lookup. Must be absent/handled per workflow policy on transitions outside on_hold.

Schema: TransitionResult (object)

Property	Type & Constraints	Req	Description
serviceItemUuid	string (uuid)	**Yes**	-
fromCode	string	**Yes**	-
toCode	string	**Yes**	-
changedAt	string (date-time)	**Yes**	-
actorUserUuid	string (uuid)	No	-
holdType	Ref: HoldType	No	-
holdReasonCode	string	No	-

Schema: HistoryEvent (object)

Property	Type & Constraints	Req	Description
fromCode	string	No	-
toCode	string	No	-
actorUserUuid	string (uuid)	No	-
occurredAt	string (date-time)	No	-
comment	string	No	-

Schema: BilledPartyChoice (object)

Eligible contacts derived from selected order doctor/default contact and scanning center. Not an unrestricted tenant contact list.

Property	Type & Constraints	Req	Description
contactUuid	string (uuid)	**Yes**	Public identifier for crm.contacts; resolve bigint ID on backend only within trusted tenant scope
role	string [enum: doctor, scanning_center]	**Yes**	Choice role in this specific order, not global contact category
label	string	**Yes**	Approved display label; avoid PHI

Schema: BilledPartyOrigin (string)

Read-only migration/user-action provenance. The unresolved_legacy state is prohibited in authoritative completed ETL.

`Allowed Enum Values: legacy_explicit, order_default, manual_resolution, user_selected, unresolved_legacy`

Schema: HoldType (string)

Hold category when placing an order item on_hold (customer vs internal).

`Allowed Enum Values: customer, internal`

Schema: HoldReasonOption (object)

Approved catalog rows once available; current demo temporary synthetic nonproduction choices must be explicitly labelled (Q-B9). Never invent actual historical reason values.

Property	Type & Constraints	Req	Description
`code`	`string`	**Yes**	Unique code in core.lookup_values (or SYNTHETIC_* in non-prod demo)
`label`	`string`	**Yes**	Human-readable label
`isTemporary`	`boolean`	**Yes**	True when option is a temporary synthetic demo choice (never production/historical authority)
`holdType`	`Ref: HoldType`	No	-
`active`	`boolean`	No	-

Schema: ZeroChargeReason (object)

Property	Type & Constraints	Req	Description
`code`	`string`	**Yes**	Unique code in core.lookup_values
`label`	`string`	**Yes**	Human-readable label (e.g., Redo / Remake, Promotional Courtesy, Voucher Full Coverage)
`description`	`string (nullable)`	No	-
`active`	`boolean`	**Yes**	-